

A common generator selected from equal convex hulls

MathlibAnnex.exists_common_of_convexHull_eq

theorem

Uses a first maximizing slice and a separating sequence to obtain an original common point carrying a prescribed property.

Statement

Let K, L be nonempty compact subsets of a finite-dimensional real normed space with $\text{conv}(K) = \text{conv}(L)$. Let η be a finite ordered list of continuous linear functionals that separates points of the ambient space. Fix another continuous linear functional ℓ_0 and a property P such that every maximizer of ℓ_0 in L satisfies P . Then there exists $z \in K \cap L$ with $P(z)$.

Assumptions

Point separation means that if $\eta(x) = \eta(y)$ for every functional in the list, then $x = y$. The property P is arbitrary; no continuity or closedness is assumed for it. It is required on every point of the maximum slice of L for ℓ_0 . Both original sets are nonempty compact, their convex hulls agree, and the ambient space is finite-dimensional over the reals.

Conclusion

The point belongs to both original generating sets and has the prescribed property. The proof obtains it in the first maximum slice of L , which is what permits the final application of the assumption on P .

For a finite real coordinate space, take the coordinate projections in any enumeration. Equality under every projection is equality of all coordinates, so this list separates points. The cited coordinate specialization supplies the same common-generator conclusion without asking the caller to construct a separating list.

Proof route

Equal convex hulls give equal maximum values and equal convex hulls after every refinement. The separating list makes a final refined set a singleton; its common convex hull then forces the same surviving original point on both sides.

▼ Proof steps

1. For any continuous linear functional ℓ , a bound on K is preserved by convex combinations. Since $K \subseteq \text{conv}(K)$ and the maximum is attained in K , this gives

$$\max_K \ell = \max_{\text{conv}(K)} \ell = \max_{\text{conv}(L)} \ell = \max_L \ell.$$

Apply this with $\ell = \ell_0$, and call the common value m_0 . Let $K_0 = \{x \in K : \ell_0(x) = m_0\}$ and $L_0 = \{x \in L : \ell_0(x) = m_0\}$. The support-face identity then gives

$$\begin{aligned} \text{conv}(K_0) &= \text{conv}(K) \cap \{x : \ell_0(x) = m_0\} \\ &= \text{conv}(L) \cap \{x : \ell_0(x) = m_0\} = \text{conv}(L_0). \end{aligned}$$

2. Write the separating list as $\eta = (\eta_1, \dots, \eta_r)$. Given nonempty compact K_{j-1}, L_{j-1} with equal convex hulls, Step 1 supplies the same maximum

$$m_j = \max_{K_{j-1}} \eta_j = \max_{L_{j-1}} \eta_j.$$

Set $K_j = \{x \in K_{j-1} : \eta_j(x) = m_j\}$ and $L_j = \{x \in L_{j-1} : \eta_j(x) = m_j\}$. These sets are nonempty compact. The support-face identity gives

$$\begin{aligned} \text{conv}(K_j) &= \text{conv}(K_{j-1}) \cap \{x : \eta_j(x) = m_j\} \\ &= \text{conv}(L_{j-1}) \cap \{x : \eta_j(x) = m_j\} = \text{conv}(L_j). \end{aligned}$$

Induction gives this equality at every stage. For $K_* = K_r$ and $L_* = L_r$ it yields

$$\text{conv}(K_*) = \text{conv}(L_*), \quad K_* \subseteq K_0 \subseteq K, \quad L_* \subseteq L_0 \subseteq L.$$

The maximum used at stage j is taken on the preceding retained sets, not on the original K, L .

3. Two survivors in L_* have equal values under every functional in the list: at the stage for a functional, both lie on that stage's maximum level, and later stages only remove points. Point separation therefore makes any two survivors equal. Since L_* is nonempty, choose $y \in L_*$ and conclude $L_* = \{y\}$. Choose $x \in K_*$. Then

$$x \in \text{conv}(K_*) = \text{conv}(L_*) = \text{conv}(\{y\}) = \{y\},$$

so $x = y$. The inclusions from the previous step show that this same point belongs to K and L .

4. Moreover $y \in L_* \subseteq L_0$, the initial maximum slice for ℓ_0 . The hypothesis on that slice yields $P(y)$. Setting $z = y$ proves the full conclusion, including the property. An empty separating list is allowed only when its separation condition holds, in which case the ambient space has at most one point and the same argument still applies.

Main citations

- Exact declaration and its source — `MathlibAnnex.exists_common_of_convexHull_eq`
- The support face generated by a maximum slice — `MathlibAnnex.convexHull_maxSlice_eq_supportFace`
- Successive refinement as a set-valued recursion — `MathlibAnnex.lexicographicRefine`
- A bound passes to the convex hull — `MathlibAnnex.le_maxValue_of_mem_convexHull`
- Equal convex hulls give equal maxima — `MathlibAnnex.maxValue_eq_of_convexHull_eq`
- A single refinement preserves equality of convex hulls — `MathlibAnnex.refine_convexHull_eq`
- Refinement stays in the original set — `MathlibAnnex.lexicographicRefine_subset`

- Survivors have equal functional values — `MathlibAnnex.apply_eq_of_mem_lexicographicRefine`
- Point separation gives at most one survivor — `MathlibAnnex.lexicographicRefine_subsingleton`
- Successive refinement preserves equality of convex hulls — `MathlibAnnex.lexicographicRefine_convexHull_eq`
- A coordinate projection — `MathlibAnnex.coordinate`
- The finite list of coordinate projections — `MathlibAnnex.allCoordinates`
- Every coordinate projection occurs in the list — `MathlibAnnex.coordinate_mem_allCoordinates`
- Coordinate projections separate points — `MathlibAnnex.allCoordinates_separate`
- The common-generator theorem in finite real coordinates — `MathlibAnnex.exists_common_pi`

Lean source signature (exact)

```
theorem exists_common_of_convexHull_eq [FiniteDimensional ℝ E]
  (ls : List (E →L[ℝ] ℝ))
  (hsep : ∀ x y : E, (∀ ℓ ∈ ls, ℓ x = ℓ y) → x = y)
  (K L : TopologicalSpace.NonemptyCompacts E)
  (hHull : convexHull ℝ (K : Set E) = convexHull ℝ (L : Set E))
  (ℓ : E →L[ℝ] ℝ)
  (Good : E → Prop)
  (hgood : ∀ z ∈ maxSlice L ℓ, Good z) :
  ∃ z, z ∈ (K : Set E) ∧ z ∈ (L : Set E) ∧ Good z
```

In the source	Mathematical meaning
<code>[FiniteDimensional ℝ E]</code>	The ambient real normed space is finite-dimensional, with its normed-space binders in the separate exact context.
<code>(ls : List (E →L[ℝ] ℝ))</code>	The finite ordered list η of continuous real-linear functionals.
<code>(hsep : ∀ x y : E, (∀ ℓ ∈ ls, ℓ x = ℓ y) → x = y)</code>	Point separation: if $\eta(x) = \eta(y)$ for every η in the list, then $x = y$.
<code>(K L : TopologicalSpace.NonemptyCompacts E)</code> <code>(hHull : convexHull ℝ (K : Set E) = convexHull ℝ (L : Set E))</code>	The original nonempty compact sets K, L satisfy $\text{conv}(K) = \text{conv}(L)$.
<code>(ℓ : E →L[ℝ] ℝ) (Good : E → Prop)</code>	The additional functional ℓ_0 and arbitrary point property P .
In the source	Mathematical meaning
<code>(hgood : ∀ z ∈ maxSlice L ℓ, Good z)</code>	The input assumption: every $z \in L$ with $\ell_0(z) = \max_L \ell_0$ satisfies $P(z)$.
<code>∃ z, z ∈ (K : Set E) ∧ z ∈ (L : Set E) ∧ Good z</code>	The conclusion supplies one and the same point $z \in K \cap L$ satisfying $P(z)$; membership is in the original sets, not just their convex hulls.

▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section

open Set
open TopologicalSpace

namespace MathlibAnnex

universe u

variable {E : Type u} [NormedAddCommGroup E] [NormedSpace ℝ E]
```

▼ Exact content identity

Declaration: MathlibAnnex.exists_common_of_convexHull_eq

Accepted content SHA-256: 186aa41ab8d1e99bcfd35e3c06411f17c94c55ab94d9e59d5ff5725fbbd783a8

Accepted source guide SHA-256: e60e678a126d7010d305842cd57f9657029df77d7b0cc816f889b35d60c37877

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73